

SPECIFICATION AND BILLS
FOR THE
SUPPLY, DELIVERY, INSTALLATION &
COMMISSIONING OF THE ELECTRICAL
INSTALLATION
AT THE
UNIVERSITY OF FORT HARE,
ALICE CAMPUS
FINANCE OFFICES CONVERSION

PREPARED BY:

CA du Toit Eastern Cape (Pty) Ltd

22 Shirley Street, Newton Park

Gqeberha (*Port Elizabeth*)

6045

CA du Toit ref No: P8080E06/FD

Contact: Mr. Petrus Strydom

CAduToit
Eastern Cape (Pty) Ltd
Consulting Engineers

December 2025

**University of Fort Hare, Alice Campus, Finance Offices Conversion,
Electrical Installation**

TABLE ON CONTENTS

1. SPECIFICATION OF THE ELECTRICAL INSTALLATION	3
1.1 STANDARD TECHNICAL SPECIFICATION	3
1.2 PROJECT SPECIFICATION	4
2. CERTIFICATES, ETC TO BE PROVIDED BY THE TENDERER	13
3. SCHEDULE OF PARTICULARS OF ELECTRICAL CONTRACTOR	14
4. PRICING INSTRUCTIONS	15
5. BILL OF QUANTITIES	18
5.1 BILLS	19
6. ANNEXURES	31
6.1 DRAWING REGISTER	31
FINAL PAGE OF THE DOCUMENT	31

1. SPECIFICATIONS OF THE ELECTRICAL INSTALLATION

1.1 GENERAL ELECTRICAL SPECIFICATION

The General technical requirements for the electrical installation will be in accordance with the Standard Electrical Specifications of the Department of Public Works. The specific documents can be downloaded from the Department's website: <http://www.publicworks.gov.za>.

The following documents shall apply:

- Section A : Preamble to the Standard Specifications
- Section B ; Installation Specifications
- Section C : Quality Specification for Materials and Equipment of Electrical Specification.

The above documents are not incorporated but nevertheless form an integral part of this specification. The contract works must be carried out in accordance with the laid down requirements. Copies of the relevant general technical specifications are available at the offices of the consulting engineer.

1.2 PROJECT ELECTRICAL SPECIFICATION

SECTION 1 - GENERAL

1 GENERAL INFORMATION

The work covered by this specification and the associated drawings must be carried out strictly in accordance with the requirements of the General Electrical Specification.

Should any requirements in the Project Specification differ from the General Electrical Specification, the requirements of the Project Specification will be given priority. Should there be any contradictions, it shall be brought to the attention of the Electrical Engineer prior to opening of tenders.

2 SITE

The site concerning this specification is the University of Fort Hare Financial Office .

3 SCOPE OF WORK

THIS INSTALLATION:

This specification covers the supply, delivery, offloading, installation, and commissioning of the Electrical Installation .

The work shall be carried out by a registered electrical contractor as per the Occupational Health and Safety Act, 1993: Electrical Installation Regulation Registration 6(4): Registration of Electrical Contractor.

The scope of work is briefly summarized as follows:

- a) Complete Electrical installation of the Electrical Installations for the proposed Additions and Alteration.
- b) All LV cable reticulation to the existing sub-Distribution Board.
- c) Install all Luminaires as per supplier's requirements to obtain full guarantee.
- d) Install trunking / wireways for reticulation of electrical power and controls.
- e) Attendance and co-ordination of all electronic services as per specification.
- f) Connection of all mechanical equipment.

4 DRAWINGS

A complete list of drawings accompanying this specification is included herein. The drawings are added as an indication of the scope of work and for information. The document must be priced as per the bills and specification.

5 MAKING GOOD

Making good to wall finishes damaged during the execution of this contract will be carried out by others, provided that the damage caused is directly associated with the forming of holes for conduits, flush boxes, or any other item of electrical equipment.

Making good unnecessary damage will be the responsibility of the electrical contractor. Any such damage will be adjudicated by the Engineer and the **Engineer's decision as to whether such damage was unavoidable or not shall be final.**

6 **WORK PROGRAM**

The electrical subcontractor must at all times liaise with the Engineer and the Main Contractor to ensure the efficient running of the contract.

The subcontractor shall take into account that there will be other subcontractors on site, and he must be prepared to work together and co-ordinate any activities with these subcontractors where necessary.

Items requiring special attention

The attention of tenderer's is directed to the following items requiring special attention:

6.1 **CO-ORDINATION OF WORK**

- a) The nature of the total building program is such that accurate co-ordination and timing of all building and contractor operations is mandatory.
- b) As such the successful tenderer (subcontractor) shall be required to work in close co-operation with the Main Contractor and other subcontractors (e.g. ceiling, ventilation, plumbing, building) to ensure that the work is properly sequenced, and to avoid any abortive work.

6.2 **SITE MEETINGS**

Regular site meetings will be held during the progress of the work. Attendance at such site meetings shall be required from time to time for all subcontractors. In the event that a representative is designated for any or all site meetings, such representative shall be empowered to make decisions on behalf of the firm, and to commit the firm to any course of action that may be decided at the meetings.

7 **RUBBLE**

The electrical subcontractor shall clear away and remove all rubble and excess materials associated with his contract from site, on a regular basis due to the fact that it is an existing facility. The site shall be entirely clean to the satisfaction of the Engineer, after completion of the contract. Care must be taken that cleaning materials are not used that can cause any damage to surfaces of walls, floors, or ceilings, etc.

8 **DELIVERY DATES**

The electrical contractor must ensure that all necessary materials are ordered and delivered in time. The delivery dates of such materials must be determined in good time. Claims for extending the contract period as a result of delays in the delivery of materials will not be considered.

9 **SECURITY AND CLOTHING**

The electrical contractor's staff shall be clearly identified and shall wear acceptable clothing. Standard coloured overalls and identification tags shall be worn at all times. The electrical contractor shall make due allowance for this in his tender price.

SECTION 2 - GENERAL

1 TESTS

After completion of the works and before first delivery is taken, a full test will be carried out on the installation for a period of sufficient duration to determine the satisfactory working thereof. During this period the installations will be inspected, and the Contractor shall make good, to the satisfaction of the Representative/Agent, any defects which may arise.

The Contractor shall provide all instruments and equipment required for testing and any water, power and fuel required for the commissioning and testing of the installations at completion.

2 MAINTENANCE OF INSTALLATIONS

With effect from the date of the Delivery the Contractor shall at his own expense undertake the regular servicing of the installation during the maintenance period and shall make all adjustments necessary for the correct operation thereof.

If during the said period the installations is not in working order for any reason for which the Contractor is responsible, or if the installations develop defects, he shall immediately upon being notified thereof take steps to remedy the defects and make any necessary adjustments.

Should such stoppages however be so frequent as to become troublesome, or should the installations otherwise prove unsatisfactory during the said period the Contractor shall, if called upon by the Representative/Agent or the Client, at his own expense replace the whole of the installations or such parts thereof as the Representative/Agent or the Client may deem necessary with apparatus specified by the Representative/Agent or the Client.

3 REGULATIONS

The installation shall be erected and tested in accordance with the Acts and Regulations as indicated in the scope of works

4 NOTICES AND FEES

The Contractor shall give all notices required by and pay all necessary fees, including any inspection fees, which may be due to the local Supply Authority.

On production of the official account, only the net amount of the fee charged by the Supply Authority for connection of the installation to the supply mains, will be refunded to the Contractor.

5 SCHEDULE OF FITTINGS

In all instances where schedule of light, socket outlet and power points are attached to or included on the drawings, these schedules are to be regarded as forming part of the specification.

6 QUALITY OF MATERIALS

Only materials of first-class quality shall be used, and all materials shall be subject to the approval of the Representative/Agent or the Client.

Wherever applicable the material is to comply with the relevant South African Bureau of Standards, specifications, or to IES Standards, where no SABS/SANS Specifications exist.

Materials wherever possible, must be of South African manufacture.

Wherever a trade name for any product has been described in the **bills of quantities / lump sum document**, the tenderer's attention is drawn to the fact that any other product of equal quality may be used subject to the written approval of the **electrical engineer** being obtained prior to the closing date for submission of tenders

If prior written approval for an alternative product is not obtained, the product described shall be deemed to have been tendered for

7 CONDUIT AND ACCESSORIES

The installation of conduits, conduit accessories and wiring shall be in strict accordance with the relevant methods as explained in this specification, the regulations of the local authority and code of practice for the wiring of premises - SANS 10142-1 - latest edition as amended.

All electronic services are to be in 25mm conduit, with separate runs for each system.

All exposed conduit is to be steel galvanised plain ended conduit.

Unless other methods of installation are specified for certain circuits, the installation shall be in conduit throughout. No open wiring in roof spaces or elsewhere will be permitted.

All conduit fittings except couplings, shall be of the inspection type. Where cast metal conduit accessories are used, these shall be of malleable iron. Zinc base fittings will not be allowed.

Bushes used for metallic conduit shall be brass and shall be provided in addition to locknuts at all points where the conduit terminates at switchboards, switch-boxes, draw-boxes, etc.

Draw-boxes are to be provided in accordance with the SANS 10142-1 and wherever necessary to facilitate easy wiring.

For light and socket outlet circuits, the conduit used shall have an external diameter of 20mm. In all other instances the sizes of conduit shall be in accordance with the SANS 10142-1 for the specified number and size of conductors, unless otherwise directed in part 2 of this specification or indicated on the drawings.

All metallic conduits shall be manufactured of mild steel with a minimum thickness of 1,2mm.

Bending and setting of conduits must be done with special bending apparatus manufactured for the purpose and which are obtainable from the manufacturers of the conduit systems. Damage to conduit resulting from the use of incorrect bending apparatus or methods applied must on indication by the Engineer, be completely removed and rectified and any wiring already drawn into such damaged conduits must be completely renewed at the Contractor's expense.

Conduit and conduit accessories used for flame-proof or explosion proof installations and for the suspension of luminaires as well as all load bearing conduit shall in all instances be of the metallic type.

7.1 CONDUIT IN ROOF SPACES

Conduit in roof spaces shall be installed parallel or at right angles to the roof members and shall be secured at intervals not exceeding 1,5m by means of saddles screwed to the roof timbers.

Nail or crampets will not be allowed.

Where non-metallic conduit has been specified for a particular service, the conduit shall be supported and fixed with saddles with a maximum spacing of 450 mm. The Contractor shall supply and install all additional supporting timbers in the roof space as required.

Under flat roofs, in false ceilings or where there is less than 0,9m of clearance, or should the ceilings be insulated with glass wool or other insulating material, the conduit shall be installed in such a manner as to allow for all wiring to be executed from below the ceilings.

Conduit runs from distribution boards shall, where possible terminate in fabricated sheet steel draw-boxes installed directly above or in close proximity to the boards.

7.2 SURFACE MOUNTED CONDUIT

Wherever possible, the conduit installation is to be concealed in the building work; however, where unavoidable or otherwise specified under Part 3 of the specification, conduit installed on the surface must be plumbed or levelled and only straight lengths shall be used.

The use of inspection bends is to be avoided and instead the conduit shall be set uniformly, and inspection coupling used where necessary.

Conduit is to be run on approved spaced saddles rigidly secured to the walls.

Alternatively, fittings, tees, boxes, couplings etc., are to be cut into the surface to allow the conduit to fit flush against the surface. Conduit is to be bedded into any wall irregularities to avoid gaps between the surface and the conduit.

Where several conduits are installed side by side, they shall be evenly spaced and grouped under one purpose-made saddle.

Distribution boards, draw-boxes, industrial switches, and socket outlets etc., shall be neatly recessed into the surface to avoid double sets. In situations where there are no ceilings the conduits are to be run along the wall plates and the beams.

Painting of surface conduit shall match the colour of the adjacent wall finishes.

Only approved plugging materials such as aluminium inserts, fibre plugs, plastic plugs, etc., and round-head screws shall be used for fixing saddles, switches, socket outlets, etc., to walls, wood plugs and the plugging in joints in brick walls are not acceptable.

7.3 CONDUIT IN CONCRETE SLABS

In order not to delay building operations the Contractor must ensure that all conduits and other electrical equipment which are to be cast in the concrete columns and slabs are installed in good time.

The Contractor shall have a representative in attendance at all times when the casting of concrete takes place. Draw-boxes, expansion joint boxes and round conduit boxes are to be provided where necessary. Sharp bends of any nature will not be allowed in concrete slabs.

Draw and/or inspection boxes shall be grouped under one common cover plate and must preferably be installed in passages or common areas.

All boxes, etc., are to be securely fixed to the shuttering to prevent displacement when concrete is cast. The conduit shall be supported and secured at regular intervals and installed as close as possible to the neutral axis of concrete slabs and/or beams.

Before any concrete slabs are cast, all conduit droppers to switchboards shall be neatly spaced and rigidly fixed.

8 WIRING

Except where otherwise specified in Part 3 of this specification, wiring shall be carried out in conduit throughout.

No wiring shall be drawn into conduit until the conduit installation has been completed and all conduit ends provided with bushes. All conduits to be clear of moisture and debris before wiring is commenced.

Unless otherwise specified in Part 3 of this specification or indicated on the service drawings, the wiring of the installation shall be carried out in accordance with the SANS10142-1. Further to the requirements concerning the installation of earth conductors to certain light points as set out in the SANS 10142-1, it is a specific requirement of this document that where plain-end metallic conduit

or non-metallic conduit has been used, earth conductors must be provided and drawn into the conduit with the main conductors to all points, including all luminaires and switches throughout the installation.

Wiring for lighting circuits is to be carried out with 1,5mm² conductors and a 1,5mm²-earth conductor. For socket outlet circuits the wiring shall comprise 2.5mm² conductors and a 2,5mm²-earth conductor. In certain instances, as will be directed in Part 2 of this specification, the sizes of the aforementioned conductors may be increased for specified circuits. Sizes of conductors to be drawn into conduit in all other instances, such as feeders to distribution boards, power points etc., shall be as specified elsewhere in this specification or indicated on the drawings. Sizes of conductors not specified must be determined in accordance with the SANS 10142-1.

The loop-in system shall be followed throughout, and no joints of any description will be permitted.

The wiring shall be done in PVC insulated 600/1000 V grade cable to SABS 150.

Where cable ends connect onto switches, luminaires etc., the end strands must be neatly and tightly twisted together and firmly secured. Cutting away of wire strands of any cable will not be allowed.

9 SWITCHES AND SOCKET OUTLETS

All switches and switch-socket outlet combination units shall be of one manufacture.

Only 16A 3 pin sockets and the new 16A RSA 3pin(similar to the Euro 2pin) are to be used, unless other special purpose types are distinctly specified or shown on the drawings.

All light switches shall be installed at 1150mm above finished floor level, to the underside of the box and all socket outlets 450mm AFFL or alternatively the height of socket outlets may be indicated on the drawings.

All light switches and socket outlets shall match the existing. All screws shall be rust proof.

10 SWITCHGEAR

For uniform appearance of switchboards, only one approved make of each of the different classes of switchgear mentioned in the Quality Specifications shall be used throughout the installations.

Only CBI Circuit breakers are to be added to the existing and new DB.

11 SWITCHBOARDS

All circuits are to be connected to the existing distribution boards.

All busbars, wiring, terminals, etc., are to be adequately insulated and all wiring is to enter the switchgear from the back of the board. The switchgear shall be mounted within the boards to give a flush front panel. Cable and boxes and other ancillary equipment must be provided where required.

Clearly engraved labels are to be mounted on or below every switch. The working of the labels in English, is to be according to the lay-out drawings or as directed by the Electrical Engineer and must be confirmed on site. Flush mounted boards to be installed with the top of the board 2,0m above the finished floor level.

12 WORKMANSHIP AND STAFF

Except in the case of electrical installations supplied by a single-phase electricity supply at the point of supply, an accredited person shall exercise general control over all electrical installation work being carried out.

The workmanship shall be of the highest grade and to the satisfaction of the Client.

All inferior work shall, on indication by the electrical engineer, immediately be removed and rectified by and at the expense of the Contractor.

13 CERTIFICATE OF COMPLIANCE

On completion of the service (practical completion), a certificate of compliance must be issued to the Electrical Engineer in terms of the Occupational Health and Safety Act, 1993 (Act 85 of 1993) and SANS 10142-1 and any other regulation pertaining to the electrical installation.

14 EARTHING OF INSTALLATION

14.1 MAIN EARTHING

The type of main earthing must be as required by the supply authority and regulations pertaining to this project.

Installations shall be effectively earthed in accordance with the SANS 10142-1 and to the requirements of the supply authority. All earth conductors shall be stranded copper with or without green PVC installation.

Connection from the main earth bar on the main board must be made to the cold water main, the incoming service earth conductor, if any and the earth mat or other local electrode by means of 12mm x 1,60 mm solid copper strapping or 16 mm² stranded (not solid) bare copper wire or such conductor as the electrical engineer may direct. Main earth copper strapping were installed below 3m from ground level, must be run in 20 mm diameter conduit securely fixed to the walls.

All other hot and cold-water pipes shall be connected with 12mm x 0,8mm perforated for solid copper strapping (not conductors) to the nearest switchboard. The strapping shall be fixed to the pipe work with brass nuts and bolts and against walls with brass screws at 150-mm centres. In all cases where metal water pipes, down pipes, flues, etc., are positioned within 1,6m of switchboards an earth connection consisting of copper strapping shall be installed between the pipe work and the board. In vertical building ducts accommodating both metal water pipes and electrical cables, all the pipes shall be earthed at each distribution board.

14.2 ROOFS, GUTTERS, AND DOWN PIPES

Where service connections consist of overhead conductors, all metal parts of roofs, gutters and down pipes shall be earthed. One bare 10mm² copper conductor shall be installed over the full length of the ceiling void, fixed to the top purlin, and connected to the main earth conductor and each switchboard. The roof and gutters shall be connected at 15m intervals to this conductor by means of 12mm X 0,8mm copper strapping (not conductors) and galvanised bolts and nuts. Self-tapping screws are not acceptable. Where service connections consist of underground supplies, the above requirements are not applicable.

14.3 SUB-CIRCUITS

The earth conductors of fall sub-circuits shall be connected to the earth busbar in the supply board in accordance with SANS 10142-1.

14.4 RING MAINS

Common earth conductors may be used where various circuits are installed in the same wire way in accordance with SANS 10142-1. In such instances the sizes of earth conductors shall be equivalent to that of the largest current carrying conductor installed in the wire way, alternatively the size of the conductor shall be as directed by the Engineer. Earth conductors for individual circuits branching from the ring main shall be connected to the common earth conductor with T-ferrules or soldered. The common earth shall not be broken.

14.5 FLEXIBLE CONDUIT

An earth conductor shall be installed in all non-metals flexible conduits. This earth conductor shall not be installed externally to the flexible conduit but within the conduit with the other conductors. The earth conductor shall be connected to the earth terminals at both ends of the circuit.

14.6 CONNECTION

Under no circumstances shall any connection points, bolts, screws, etc., used for earthing be utilised for any other purpose. It will be the responsibility of the Contractor to supply and fit earth terminals or clamps on equipment and materials that must be earthed where these are not provided. Unless earth conductors are connected to proper terminals, the end shall be tinned and lugged.

15 MOUNTING AND POSITIONING OF LUMINAIRES

The Contractor is to note that in the case of board and acoustic tile ceilings, i.e., as opposed to concrete slabs, close co-operation with the building contractor is necessary to ensure that as far as possible the luminaires are symmetrically positioned with regard to the ceiling pattern.

The layout of the luminaires as indicated on the drawings must be adhered to as far as possible and must be confirmed with the electrical engineer.

Earth conductors must be drawn in with the circuit wiring and connected to the earthing terminal of all fluorescent luminaires as well as other luminaires exposed to the weather in accordance with the SANS 10142-1.

SECTION 3 – INSTALLATION DETAILS

1 CABLE SLEEVE PIPES

Where cables cross under roadways, other services and where cables enter buildings, the cables shall be installed in asbestos-cement pipes, earthenware, or high-density polyethylene pipes.

The ends of all sleeves shall be sealed with a non-hardening watertight compound after the installation of cables. All sleeves intended for future use shall likewise be sealed.

2 NOTICES

The Contractor shall issue all notices and make the necessary arrangements with Supply Authorities, the Postmaster-General, S.A. Transport Services, Provincial or National Road Authorities and other authorities as may be required with respect to the installation.

3 ELECTRICAL EQUIPMENT

All equipment and fittings supplied must be in accordance with the specification, suitable for the relevant supply voltage and frequency and must be approved by the Lactalis' representative.

4 DRAWINGS

The drawings generally show the scope and extent of the proposed work and shall not be held as showing every minute detail of the work to be executed.

The position of power points, switches and light points that may be influenced by built-in furniture must be established on site, prior to these items being built in.

5 BALANCING OF LOAD

The Contractor is required to balance the load as equally as possible over the multiphase supply.

6 SERVICE CONDITIONS

All plant shall be designed for the climatic conditions appertaining to the service.

7 SUPPLY VOLTAGE

The electrical supply will be from an existing distribution boards of the facility, which 400V/3PH.

8 DISTRIBUTION BOARDS

8.1 GENERAL

The equipment is to be Schneider manufacturers.

The contractor shall check the size and the position of the switchboards on site before manufacture is commenced.

The contractor shall also submit a copy of the manufacturer's working drawings for the switchboard to the Engineer for written approval before manufacture is commenced.

The contractor shall inform the Engineer in good time of the completion of the switchboards by the manufacturer. The switchboards shall be inspected and approved by the Engineer at the manufacturer's factory before such switchboard is delivered to site. Only registered switchboard manufactures within the Port Elizabeth area are to be used, due to the above requirements.

The tenderer's attention is drawn to the fact that any other supplier/manufacturer, of equal quality, may be used, subject to the written approval of the **Electrical Engineer** being obtained prior to the closing date for submission of tenders. If prior written approval for an alternative

supplier/manufacture is not obtained, the preferred supplier/manufacture shall be deemed to have been used for the tender.

Wherever a trade name for any product has been described in the **bills of quantities / lump sum document**, the tenderer's attention is drawn to the fact that any other product of equal quality may be used subject to the written approval of the **Electrical Engineer** being obtained prior to the closing date for submission of tenders. If prior written approval for an alternative product is not obtained, the product described shall be deemed to have been tendered for

The switchboards shall only be accessible from the front. All equipment and wiring shall therefore only be accessible after the removal of the face plates. All equipment shall be installed flush with the switchboard face plates and of the DIN rail type in accordance with IEC-60715. The equipment is to be Schneider manufacturers.

Switchboards must be fitted with appropriate lock out mechanisms.

8.2 LABELLING OF SWITCHBOARD

All switchgear labels on the face plates of the switchboard shall be fitted with permanently engraved labels black-on-white, securely fixed to the distribution board.

Each single way on the switchboard shall be numbered consecutively. The switchboard is to have a circuit legend card typewritten and displayed inside the hinged door of the switchboard behind a clear perspex sheet in a fixed card holder.

All label wording shall be confirmed with the Engineer before the title cards are typed. (Please note that handwritten legend cards are unacceptable). The following descriptions must appear on the various title cards.

- a) Way number
- b) Circuit number
- c) Switchboard number
- d) Description, e.g., Main switch or lights or plugs, etc.

8.3 EQUIPMENT ON SWITCHBOARDS

All equipment used, must be able to withstand a minimum fault current of 5/6kA, or as otherwise specified, without causing damage to such equipment. The contractor shall be responsible for the phasing of all single phase 240V circuits to result in a balanced 3-phase system.

- All equipment installed in the distribution boards is to be either SABS or IEC approved.
- There must be at least 40% spare mountable space in each of the distribution boards.
- There must be enough clear space to terminate and gland the incoming cables, as well as any outgoing cables.
- All distribution boards are to be manufactured within accordance of the relevant SANS regulations.
- There must be space on the top and sides to be able to easily wire all the circuits in the distribution boards.
- Any busbars used, must have the corrected depreciation rating, if been used in parallel.

9 INSTALLATION OF LV CABLES

Supply and install the following PVC/PVC/SWA/ECC/PVC 600/100V cables to SABS 150 as amended and as per quantities in the bill.

10 TRUNKING AND WIRE BASKET INSTALLATION

The installation of trunking, trunking accessories and wiring shall be in strict accordance with the relevant methods as explained of the local authority and code of practice for the wiring of premises, SANS 10142, latest edition as amended.

The trunking for the power and lighting is to be P8000 trunking, linking the DB's to the various areas. Where there are ceilings, all power circuit wiring, for lighting installation, is to run in the P8000.

All ICT services are to run in 150mm wire baskets or cable trays, heavy duty, installed in the ceiling spaces. Where there are no ceilings the 25mm conduits are to be installed in the slabs and are to be concealed. The wire basket is to be properly supported as per the requirements of the manufacturer. All connections and bends are to be done with the appropriate junctions and bends supplied by the manufacturer.

11 OUTLETS

11.1 SOCKET OUTLETS

All switch socket outlets, not in the power skirting, shall be connected with 20mm conduit with 100 x 100 x 50 mm flush boxes wherever possible and shall be flush mounted in the walls as indicated on the drawings.

All switch socket outlets shall be installed at 450mm AFFL unless otherwise indicated on the power layout drawing. Heights above finished floor level shall be to the underside of the outlet box.

All socket outlets shall match the existing. All screws shall be rust proof.

All switch socket outlets shall be wired with 2.5 mm², single core PVC insulated copper conductors throughout with a 2,5 mm² green/yellow PVC insulated copper earth wire to each circuit.

11.2 MECHANICAL EQUIPMENT

The electrical sub-contractor shall supply, deliver, and install the isolator for the connection of fans, air-conditioners and other mechanical equipment, as per the drawings. It is the responsibility of the Mechanical contractor to connect from the isolator to his equipment and to make the correct voltage and phase rotation of the equipment been connected.

12 LIGHTING INSTALLATION

Shall be an all-conduit installation with not less than 20-mm dia. (SABS) approved conduit being installed.

Wiring shall be not less than 1,5 mm², single core PVC insulated copper conductors throughout with 1,5 mm² green coloured/yellow PVC insulated earth wire.

Before the luminaires are installed, the electrical subcontractor shall ensure that the said items are properly cleaned and that all metal work is properly finished without scratches or other soiling marks.

The electrical subcontractor shall supply, install, and commission the light fittings in the positions indicated on the drawings.

12.1 LIGHT SWITCHES

Flush mounted switches shall be suitable for a load of 16A. Mounting heights shall be at 1150mm above finished floor level to the underside of the box, unless otherwise indicated on the drawings and denotes the height from finished floor level to the lower lip of the outlet box.

All light switches shall be of the Onesto, Matrix Range, manufacture with the black inserts and silver cover. All screws shall be rust proof.

The electrical subcontractor shall submit samples and obtain final approval of the cover plates and switches before ordering for the installation to enable any changes to be made.

12.2 PHOTOCELL SENSORS

The photocell (day/night) sensors are to be surface mounted as indicated on the drawings. They are to control the outside lighting of the buildings, and some are connected via a contractor, see distribution board drawings. The type of photocell is to be a square unit with various inlet options and to have an opal lense with off-white/grey base, **Spectrum Pecu QS16T**.

12.3 LUMINAIRES

All the internal wiring of the luminaires must conform to the SANS/IEC-60598 compulsory safety specification. The guarantee period offered by the manufacture of the electronic drivers and LED units must be at least 5 years.

The light fittings to each of the areas of the installation shall be as indicated on the ceiling layout drawing for each area. It is to be noted that the spacing and position of the luminaires on the drawings are indicative and exact positioning of luminaires must be determined on site.

All lighting circuits shall be wired with 2,5mm², single core PVC insulated copper conductors throughout with a 2,5mm² green/yellow PVC insulated copper earth wire to each circuit.

12.4 SCHEDULE OF LUMINAIRES

Luminaires must be according to the specification for tender purposes, so that all tender prices are based on the exact same fitting. Listed below is the list of already approved luminaires. Listed below is the list of already approved luminaires from the local manufacturer Lighting Innovations as preferred by the client.

Luminaire Schedule:

Type	Description	Picture
A	Recessed, 600x600, back-lit LED, ±40W and ±5100lumen, 4000k, CRI >80Ra, complete with control gear and 5A plug & cable.	
B	Bulkhead, round, black trim, aluminium base, LM6 for coastal conditions, s/steel screws, IP66, ±15W, ±2098lumen, 4000k, complete with integral control gear.	
C	55mm wide x 80mm high Linear Light, 4000k, 30W/m, 6309lm, suspended from ceiling. Colour by Architect	
D	Downlight, recessed, LED, GU10, 4000K, Low Glare, Spazio 2224, White.	
E	Footlight, ±2W to fit standard 100x100 box, 3000K Black	

F	Versatile Linear Fitting, LED, ±71W, ±6358lumen, ± 1200mm Diameter, 4000k, suspended, CRI >80Ra, complete with suspension kit and internal control gear.	
G	A small inground up light GU10 lamp with a die cast aluminium trim, suitable for illuminating columns, wall accents, architectural and landscape features.	
H	Bulkhead, rectangle, 45deg, black, 24W coastal conditions	
T	Splash/dust proof LED fitting, with integral LED controller, ±51W, ±7660lumen, 4000k, ±1277mm, CRI >80Ra, min IP65	

13 **CONDUIT INSTALLATION**

The installation of conduits, conduit accessories and wiring shall be in strict accordance with the relevant methods as explained in this specification, the regulations of the local authority and code of practice for the wiring of premises - SANS 10142-1 - latest edition as amended.

All electronic services are to be in 25mm conduit, with separate runs for each system.

14 **ELECTRONIC SERVICES**

Liaison with the various ICT services with their installation and making sure all their power requirements and wireways are installed correctly and to their requirements.

The systems equipment is as per the specific items in the bill and quantities and installation as per the manufacturer's specification and the general standard departmental specifications.

15 **MONEY PROVISION FOR SPECIFIED ITEMS**

Each money provision is to be used only for the relevant specified item and deducted in whole or in part if not required. The work covered by each money provision shall be measured and valued by the Engineer after the receipt of invoices, receipts of payment etc. from the electrical subcontractor.

16 **EARTHING OF THE INSTALLATION**

16.1 **GENERAL**

The installation must be properly earthed to comply with the requirements of SANS 10142-1 - latest edition as amended: Code of Practice for the Wiring of Premises, and according to the by-laws of the local Supply Authority..

16.2 **EARTH CONNECTIONS**

- a) All individual components shall be connected to the earth connection of the domestic earth supply, or according to the requirements of the Project Specification.
- b) Under no circumstances shall the latter domestic earth supply be interconnected to the earth supply utilised for electronic control equipment and installations.

16.3 **EARTHING OF SUB-CIRCUITS AND COMPONENTS**

- a) Earth conductors of all sub-circuits shall be connected to the earth bar of the main earth connection as provided.
- b) The ends of all metal channels, cable trays, etc., containing cables and conductors, shall be earthed to the nearest distribution board by means of earth straps or conductors.
- c) The latter earth connections shall be connected to the domestic earth supply system.
- d) All metal conduits shall, where installed by the Contractor, be terminated at control boards or junction boxes. Where this cannot be done, conduit ends shall be earthed separately by means of stranded earth conductors according to the requirements of the regulations.
- e) Earth conductors shall be installed in all flexible type of conduit to interconnect the two ends of the flexible conduit.

17 **DOCUMENTATION**

17.1 **GENERAL REQUIREMENTS**

- a) The Contractor shall prepare workshop drawings indicating detail of all components and the composition of the installation. The drawings must be prepared in such a way that sufficient information and detail is shown in order that all equipment and components of the installation can be manufactured.
- b) Workshop drawings of the equipment shall be submitted to the Engineer for approval before manufacturing and/or installation of components commences.
- c) In the event that there is a deviation from the equipment specified, or there is a deviation of the equipment layout from the general layout as indicated on the drawings, such deviations shall be approved by the Engineer before the equipment is manufactured.

17.2 **MODIFICATIONS TO DRAWINGS**

- a) After the Engineer has, where necessary, revised, or modified drawings which had been submitted for approval, the Contractor shall supply to the Engineer one (1) copy of all revised and modified drawings as explained above.
- b) In the event that drawings are revised or modified during the construction phase, a copy of a drawing indicating the modifications shall again be supplied to the Engineer.
- c) Irrespective of the abovementioned copies, the Contractor or his representative shall keep one (1) copy of each drawing on site. Such drawings shall be made available to the Engineer on request on site.
- d) All final drawings shall be certified as correct by the Contractor after the installation has been completed.

18 **FINAL DRAWINGS, OPERATOR'S, AND MAINTENANCE MANUALS**

18.1 **GENERAL**

- a) All manuals shall be bound properly and thoroughly in a durable hard plastic cover. The contents of the manual shall be clearly legible, well-structured and supplied with an index.

- b) No Completion Certificate shall be issued before all drawings, two copies of the operator's manuals and two copies of the maintenance manuals have been supplied to and approved by the Engineer.
- c) Concept issues of final drawings and manuals shall be ready and available with the first take over and commissioning and no take over shall be considered without such drawings and manuals.

2. CERTIFICATES, ETC TO BE PROVIDED BY THE TENDERER

The following Returnable Certificate's, references, etc of the Electrical contractor shall be attached to this page:

- A valid Tax Clearance Certificate issued by the South African Revenue Services.
- A copy of the Certificate of Incorporation of the company / close corporation / partnership. In the case of a joint venture between 2 or more firms, the tenderer shall attach a copy of the documents of incorporation of the joint venture.
- Registration certificate as an electrical contractor tendering, from the Department of Labour, as per the Occupational Health and Safety Act, 1993: Electrical Installation Regulation Registration 6(4): Registration of Electrical Contractor.
- Proof of employment of the registered person, as mentioned on the above registration certificate, if not the owner, director or shareholder of the company tendering.
- Company profile, showing experience in the electrical contracting field and particular reference to similar projects of this size and nature.
- Letter of Good Standing with the Compensation Commissioner.
- Proof of Registration with N.B.C.E.I. and their relevant employers.

At tender submission, only copies are required. originals and certified copies will be required from the successful tenderer for the contract.

3. SCHEDULE OF PARTICULARS OF ELECTRICAL CONTRACTOR

This schedule must be completed and submitted together with the bid. Should this requirement not be complied with in full, the bid may be considered invalid.

Name of Electrical Contractor: _____
As on the registration certificate as an electrical Contractor, from the Department of Labour, as per the Occupational Health and Safety Act, 1993: Electrical Installation Regulation Registration 6(4): Registration of Electrical Contractor.

Physical Address: _____



Contact Person: _____

Cell Phone Number: _____

E-Mail Address: _____

As on the registration certificate as an electrical contractor, from the Department of Labour, as per the Occupational Health and Safety Act, 1993: Electrical Installation Regulation Registration 6(4): Registration of Electrical Contractor.

Registered Number: _____

Name of installation electrician on registration certificate: _____

Registration number of the above person: _____

Name of Electrical Contractor:

Signature of Electrical Contractor:

Date:

4. PRICING INSTRUCTIONS

2.1.1 PREAMBLE TO THE SCHEDULE OF QUANTITIES

- 2.1.1.1 Unless otherwise stated, items are measured nett in accordance with the drawings, and no allowance is made for waste.
- 2.1.1.2 The quantities set out in the Schedule of Quantities are the estimated quantities of the Contract Works, but the Contractor will be required to undertake whatever quantities may be directed by the Engineer from time to time. The Contract Price for the completed contract shall be computed from the actual quantities of work done, valued at the relevant unit rates and prices.
- 2.1.1.3 The prices and rates to be inserted in the Schedule of Quantities are to be the full inclusive prices for the work described under the several items. Such prices and rates shall cover all costs and expenses that may be required in and for the execution of the work described, and shall cover the cost of all general risks, liabilities, and obligations set forth or implied in the documents on which the tender is based, as well as overhead charges and profit. Reasonable prices shall be inserted as these will be used as a basis for assessment of payment for additional work that may have to be carried out.
- 2.1.1.4 A price or rate is to be entered against each item in the Schedule of Quantities, whether the quantities are stated or not. An item against which no price is entered will be considered to be covered by the other prices or rates in the Schedule.
- 2.1.1.5 Except where rates only are required, the Tenderer shall insert all amounts to be included in his total tendered price in the "Amount" column and show the corresponding total tendered price.
- 2.1.1.6 All prices or rates inserted in the Schedule of Quantities shall be EXCLUDING VAT. Provision has been made on the Summary Page, of the Schedule of Quantities, for the addition of VAT.
- 2.1.1.7 Arithmetical errors of responsive tenders will be corrected in the following manner:
- Where there is a discrepancy between an amount shown in figures, and the corresponding amount stated in words, the amount stated in words shall take preference.
 - In the Schedule of Quantities, if there is an error in the line item total resulting from the product of the quantity and the unit rate, the line item total shall govern, and the rate shall be corrected.
 - Where there is a misplacement of the decimal point in the unit rate, the line item total shall govern and the unit rate will be corrected.
 - Where there is an error in the total of the prices either as a result of other corrections required by this checking process or in the tenderer's addition of prices, the total of the prices shall govern and the tenderer will be asked to revise selected item prices (and their rates in the Schedule of Quantities) to achieve the tendered total of the prices.

Should a tenderer be unwilling to make the corrections ordered by the Engineer, the tender may be disqualified.

- 2.1.1.8 The units of measurement described in the Schedule of Quantities are metric units. Abbreviations used in the Schedule of Quantities are as follows :

mm	=	millimetre	hrs	=	hours
m	=	metre	kg	=	kilogram
km	=	kilometre	t	=	ton (1 000 kg)
m ²	=	square metre	No.	=	number
m ² .pass	=	square metre-pass	sum	=	lump sum
ha	=	hectare	MN	=	MegaNewton
m ³	=	cubic metre	MN.m	=	MegaNewton-metre
m ³ .km	=	cubic metre-kilometre	P C sum	=	Prime Cost sum
ℓ	=	litre	Prov sum	=	Provisional sum
kℓ	=	kilolitre	%	=	per cent
MPa	=	MegaPascal	kW	=	kilowatt

2.1.2 PRICING OF PRELIMINARIES

- 2.1.2.1 Preliminary and general requirements are based on the various parts of SANS 1921, Construction and management requirements for works contracts. The additions, deletions and alterations to the various parts of SANS 1921 as well as the contract specific variables are as stated in the Specification Data in the Scope of Work. Only the headings and clause numbers for which allowance must be made in the Bills of Quantities are recited.
- 2.1.2.2 It will be assumed that prices included in the Bills of Quantities are based on Acts, Ordinances, Regulations, By-laws, International Standards and National Standards that were published 28 days before the closing date for tenders. (Refer to www.stansa.org.za or www.iso.org for information on standards).
- 2.1.2.3 The drawings listed in the Annexure used for the setting up of these Bills of Quantities are kept by the Principal Agent or Agent and can be viewed at any time during office hours up until the completion of the works.
- 2.1.2.4 Reference to any particular trademark, name, patent, design, type, specific origin or producer is purely to establish a standard for requirements. The offer may use others which must be approved in writing, by the engineer, prior to the closing date of the tender. No other alternatives will be accepted after the closing date.
- 2.1.2.5 Where any item is not relevant to this specific contract, such item is marked N/A (signifying “not applicable”)
- 2.1.2.6 The Contract Data and the standard form of contract referenced therein must be studied for the full extent and meaning of each and every clause set out in Section 1 (Preliminaries) of the Bills of Quantities
- 2.1.2.7 The Bills of Quantities is not intended for the ordering of materials. Any ordering of materials, based on the Bills of Quantities, is at the Contractor’s risk.
- 2.1.2.8 The amount of the Preliminaries to be included in each monthly payment certificate shall be calculated from the priced preliminaries items in the bills of quantities document. The contractor and the contractor shall agree on a division of the priced preliminaries items into:
- an initial or establishment charge
 - a monthly charge
 - a final or disestablishment charge

Where the initial construction period is extended, the monthly charge shall be recalculated on the same basis as was originally applied but taking into account the revised construction period and the amounts already paid to the contractor. Should the contractor and the contractor be unable to agree such division then the principal agent shall make a division

of the amount of preliminaries to be incorporated in the valuations for each monthly payment certificate.

- 2.1.2.9 The amount or items of the Preliminaries shall be adjusted to take account of the theoretical financial effect which changes in time or value (or both) have on this section. Such adjustments shall be based on adjustments in the following categories as recorded in the Bills of Quantities:
- a) An amount which is not to be varied, namely Fixed (F)
 - b) an amount which is to be varied in proportion to the contract value as compared with the contract sum, namely Value Related (V); and
 - c) an amount which is to be varied in proportion to the construction period as compared to the initial construction period excluding revisions to the construction period for which the contractor is not entitled to adjustment of the contract value in terms of the agreement, namely Time Related (T).
- 2.1.2.10 Where no provision is made in the Bills of Quantities to indicate which of the three categories in 11 apply or where no selection is made, the adjustments shall be based on the following breakdown:
- a) 10 percent is Fixed;
 - b) 15 percent is Value Related
 - c) 75 percent is Time Related.
- 2.1.2.11 The adjustment of the Preliminaries shall apply notwithstanding the actual employment of resources in the execution of the works. The contract value and contract sum used for the adjustment of the Preliminaries shall exclude any contingency sum, the amount for the Preliminaries and any amount in respect of contract price adjustment provided for in the contract. Adjustments in respect of any staged or sectional completion shall be prorated to the value of each section.

5. BILL OF QUANTITIES

BILLS OF QUANTITIES GENERAL NOTES

The Tender Price must exclude VAT. All rates, provisional sums, etc. in these Bills of Quantities must be nett with VAT calculated and added to the total value thereof in the Final Summary.

1. This Bill of Quantities form part of and must be read in conjunction with the Specifications and must be submitted, duly completed, on the closing date of tenders.
2. Tenderer's must complete the Bill of Quantities and detail the unit rate and total amount of each item. The "Total" shall constitute the tender price for adjudication.

NOTE:

Tenderers are advised to check their item extensions and total additions as arithmetical errors occurring in the priced Bill of Quantities cannot be considered.

3. No alterations, erasure or addition is to be made in the text of the Bill of Quantities. Should any alteration, erasure or addition be made it will not be recognised but the original wording of the Bills of Quantities will be adhered to.
4. The Engineer will check the completed Bill of Quantities and reserves the right to adjust any individual price and to rectify any discrepancy whilst the total tender price as quoted remains unaltered.
5. The quantities given in the Bills for cannot be regarded as exact as this is a provisional bill and will be re-measurement on site after completion of the service and adjustments will be made according to the unit rates given in the Bill

CHECKING OF CABLE LENGTHS:

Notwithstanding the fact that the length of cables as given in the Bill of Quantities have been measured from scaled drawings, the Contractor shall check such lengths on site before ordering the cable as he will not be paid for excess cable after the completion of the service. Any allowance for off-cuts shall be made in the unit rate. The final measurement shall be based on the nett route length of the cables concerned.

6. Where alternative prices for equipment of different manufacture is quoted the lowest alternative price for equipment to the specification must be quoted against the relevant item in the Bill of Quantities. The remaining alternative price must be furnished separately.
7. The unit prices quoted in the Bill of Quantities must include for such small installation materials as are required for the complete installation in accordance with the Specification.

5.1 BILLS

BILL: ELECTRICAL INSTALLATION

Item	Description	Unit	Qty	Rate	Amount	
					R	C
1	Preliminary and General Items, pertaining specifically to the Electrical Installation.					
1.1	Nett price for site establishment and compliance with the conditions of contract, including storage, insurance and other overheads.	Sum	1	R	R	
1.2	Insurance and Guarantee	Sum	1	R	R	
1.3	Allowance for liaising with the Technical Department, for switching off of the power and identifying the local distribution boards.	Sum	1	R	R	
1.4	Liaison with Main Contractor and for attendance for the checking of Builders' and other Sub-contractors' work pertaining to the Electrical Installation.	Sum	1	R	R	
1.5	Liaison with the various ICT services with their installation and making sure all their power requirements and wireways are installed correctly. Namely, CCTV, Data, Access Control, etc.	Sum	1	R	R	
1.6	Nett price to comply with all the requirements of the Health & Safety specifications of the main contractor.	Sum	1	R	R	
1.7	Site supervision and reporting according to the Health and Safety Specifications.	Sum	1	R	R	
1.8	Supplying or hiring, erection & dismantling of complete mobile cranes, scissor jacks and cherry pickers, including access ladders, warning and safety signs as required for the installation for the duration of the contract.	Sum	1	R	R	
1.9	All labour, testing and issuing of the appropriate COC's. with regards to the Electrical Installation	Sum	1	R	R	
1.10	Compilation of detail Operating and Maintenance Manuals, as-built drawings and documentation as specified and to the satisfaction of the Engineer. 1-off printed copy and 2 x electronic versions on memory stick.	Sum	1	R	R	
Sub total for Section1 - Preliminary and General Items					R	

BILL: ELECTRICAL INSTALLATION

Item	Description	Unit	Qty	Rate	Amount
Brought Forward from previous Page					R
2.3	Distribution Boards Supply and Install in position the distribution boards, complete with trays, labels, conduit and trunking terminations as required and all other necessary items as per the drawing, P8080E01-003.				
	Supply				
2.3.1	DB-FO	No.	1	R	R
	Install				
2.3.2	DB-FO	No.	1	R	R
2.3.3	Add the following equipment complete to the existing DB: 80 TP CB, 10kA.	Sum	1	R	R
2.4	P9000 Trunking Supply and install the P9000 trunking complete with hangers brackets, covers and threaded rods, purpose made bends and offsets as indicated on the drawings.				
	Supply				
2.4.1	P9000 trunking, complete with splices and pvc covers.	m	45	R	R
2.4.2	End cap for P9000	No.	2	R	R
2.4.3	Elbow, 90° bend for P9000	No.	1	R	R
2.4.4	T-Piece for P9000	No.	1	R	R
2.4.5	Internal or external riser for P9000	No.	1	R	R
2.4.6	Distribution outlet for P9000	No.	1	R	R
	Install				
2.4.7	P9000 trunking, complete with splices and pvc covers.	m	45	R	R
2.4.8	End cap for P9000	No.	2	R	R
2.4.9	Elbow, 90° bend for P9000	No.	1	R	R
2.4.10	T-Piece for P9000	No.	1	R	R
2.4.11	Internal or external riser for P9000	No.	1	R	R
2.4.12	Distribution outlet for P9000	No.	1	R	R
Carried Forward to next Page					R

BILL: ELECTRICAL INSTALLATION

Item	Description	Unit	Qty	Rate	Amount
Brought Forward from previous Page					R
2.5	Welded Wire Mesh medium duty cable tray Supply and install brackets, cross arms and Welded Wire Mesh galvanised medium duty wire cable tray complete with hangers brackets, covers and threaded rods, couplers etc.				
	Supply				
2.5.1	Welded Wire Mesh MD cable tray - 150mm wide	m	45	R	R
2.5.2	Welded wire mesh 90° Horizontal Elbow, internal or external , 150mm	No.	2	R	R
2.5.3	Tee Piece - 150mm	No.	1	R	R
	Install				
2.5.4	Welded Wire Mesh MD cable tray - 150mm wide	m	45	R	R
2.5.5	Welded wire mesh 90° Horizontal Elbow, internal or external , 150mm	No.	2	R	R
2.5.6	Tee Piece - 150mm	No.	1	R	R
2.6	PVC Conduit and PVC trunking PVC conduit and accessories. Conduit applied and installed, including couplings, cutting, reaming, bending, fixing, etc. as per specification.				
	Supply				
2.6.1	20 mm diameter	m	572	R	R
2.6.2	25 mm diameter	m	77	R	R
2.6.3	20 mm diameter - Flexible conduit (Sprague)	m	34	R	R
	Install				
2.6.4	20 mm diameter	m	572	R	R
2.6.5	25 mm diameter	m	77	R	R
2.6.6	20 mm diameter - Flexible conduit (Sprague)	m	34	R	R
2.7	Conduit boxes for PVC Conduit Outlet boxes, round 65mm diameter x 25mm deep, complete with covers and screws, with conduit ends .				
	Supply				
2.7.1	Round box and lid for 20mm conduit	No.	109	R	R
2.7.2	Round box and lid for 25mm conduit	No.	14	R	R
Carried Forward to next Page					R

BILL: ELECTRICAL INSTALLATION

Item	Description	Unit	Qty	Rate	Amount
Brought Forward from previous Page					R
2.7.3	100 x 50 x 50 Box - Recessed or Surface	No.	10	R	R
2.7.4	100 x 100 x 50 Box - Recessed or Surface	No.	20	R	R
Install					
2.7.5	Round box and lid for 20mm conduit	No.	109	R	R
2.7.6	Round box and lid for 25mm conduit	No.	14	R	R
2.7.7	100 x 50 x 50 Box - Recessed or Surface	No.	10	R	R
2.7.8	100 x 100 x 50 Box - Recessed or Surface	No.	20	R	R
2.8 Conductors					
PVC insulated 600/1000V Copper conductors, including earth, drawn into circuit.					
Supply					
2.8.1	1.5 mm ² - (House Wire) (Supply - continued)	m	3 089	R	R
2.8.2	2.5 mm ² - (House Wire)	m	2 402	R	R
2.8.3	4 mm ² - (House Wire)	m	400	R	R
Install					
2.8.4	1.5 mm ² - (House Wire)	m	3 089	R	R
2.8.5	2.5 mm ² - (House Wire)	m	2 402	R	R
2.8.6	4 mm ² - (House Wire)	m	400	R	R
2.9 Power Skirting					
Power skirting, OBO, white, 2 tier, complete with covers, end caps and all accessories.					
Supply					
2.9.1	2 tier Power Skirting, OBO White, complete with covers, splices etc.	m	40	R	R
2.9.2	16A SSO, 164-1, skirting type complete with covers and cradles (Normal)	No	1	R	R
2.9.3	16A, SSO, 164-2 (Euro) with switch skirting type complete with, switch, covers and cradles (Normal)	No.	1	R	R
2.9.4	CP2, Set of outlets in power skirting consisting of: 1xold RSA SSO, 1xnew RSA SSO(Euro), 2xDed SSO, 1xUSB SSO, 2xRJ45 outlets	No.	24	R	R
Carried Forward to next Page					R

BILL: ELECTRICAL INSTALLATION

Item	Description	Unit	Qty	Rate	Amount
Brought Forward from previous Page					R
2.9.5	Data/Telephone outlets, 1 x RJ45 complete with cradle & covers, skirting type	No	1	R	R
2.9.6	USB Duo & blank, skirting type complete with covers and cradles	No.	1	R	R
2.9.7	Internal Corners for Power Skirting - 2 Tier	No.	1	R	R
2.9.8	End caps for Power Skirting - 2 Tier	No	16	R	R
2.9.9	Install 2 tier Power Skirting, OBO White, complete with covers, splices etc.	m	40	R	R
2.9.10	16A SSO, 164-1, skirting type complete with covers and cradles (Normal)	No	1	R	R
2.9.11	16A, SSO, 164-2 (Euro) with switch skirting type complete with, switch, covers and cradles (Normal)	No.	1	R	R
2.9.12	CP2, Set of outlets in power skirting consisting of: 1xold RSA SSO, 1xnew RSA SSO(Euro), 2xDed SSO, 1xUSB SSO, 2xRJ45 outlets	No.	24	R	R
2.9.13	Data/Telephone outlets, 1 x RJ45 complete with cradle & covers, skirting type	No	1	R	R
2.9.14	USB Duo & blank, skirting type complete with covers and cradles	No.	1	R	R
2.9.15	Internal Corners for Power Skirting - 2 Tier	No.	1	R	R
2.9.16	End caps for Power Skirting - 2 Tier	No	16	R	R
2.10	Switched socket outlets Supply, install and connect recessed 16 A, switched socket outlet complete with cover plate, 100x100.				
	Supply				
2.10.1	Double Switched Socket Outlet, 100 x 100 (4x4) 1 x std old RSA SSO 164-1 & 1 x new std RSA SSO 164-2.	No.	13	R	R
2.10.2	Double Switched Socket Outlet - 100x100, dedicated, shaved earth, red.	No.	1	R	R
2.10.3	Single SSO - Weather proof - S15	No.	1	R	R
Carried Forward to next Page					R

BILL: ELECTRICAL INSTALLATION

Item	Description	Unit	Qty	Rate	Amount
Brought Forward from previous Page					R
2.10.4	Three Compartment Floor pop-up box complete equipped as per CP2	No.	3	R	R
2.10.5	Double RJ45 outlet, 100x50,	No.	2	R	R
2.10.6	HDMI outlet, 100x50,	No.	2	R	R
2.10.7	Blank 100x100 cover for mech wiring/controls	No.	4	R	R
2.10.8	Install Double Switched Socket Outlet, 100 x 100 (4x4) 1 x std old RSA SSO 164-1 & 1 x new std RSA SSO 164-2.	No.	13	R	R
2.10.9	Double Switched Socket Outlet - 100x100, dedicated, shaved earth, red.	No.	1	R	R
2.10.10	Single SSO - Weather proof - S15	No.	1	R	R
2.10.11	Three Compartment Floor pop-up box complete equipped as per CP2	No.	3	R	R
2.10.12	Double RJ45 outlet, 100x50,	No.	2	R	R
2.10.13	HDMI outlet, 100x50,	No.	2	R	R
2.10.14	Blank 100x100 cover for mech wiring/controls	No.	4	R	R
2.11	Isolators				
	Supply				
2.11.1	20 A double pole isolator complete, flush or surface, 100x50.	No.	1	R	R
2.11.2	Weather Proof, Surface 32A double pole isolator complete.	No.	3	R	R
2.11.3	Weather Proof, Surface 63A triple pole isolator complete.	No.	1	R	R
2.11.4	Install 20 A double pole isolator complete, flush or surface, 100x50.	No.	1	R	R
2.11.5	Weather Proof, Surface 32A double pole isolator complete.	No.	3	R	R
2.11.6	Weather Proof, Surface 32A triple pole isolator complete.	No.	4	R	R
Carried Forward to next Page					R

BILL: ELECTRICAL INSTALLATION

Item	Description	Unit	Qty	Rate	Amount
Brought Forward from previous Page					R
2.12	Switching units				
	Supply				
2.12.1	16 A , 1-lever , 1-way - 50x100,	No.	8	R	R
2.12.2	16 A , 1-lever , 1x2-way - 50x100,	No.	2	R	R
2.12.3	16 A , 1-lever and Rotary dimmer - Shuttle - 50x100,	No.	1	R	R
2.12.4	16 A , 1-lever, 1x1way, Weatherproof - IP65	No.	1	R	R
2.12.5	Recessed single-load 360deg PIR occupancy sensor. Schneider Electric: SAE-UE-MS-CSBWE	No.	14	R	R
2.12.6	Photocell, 16A, Daylight Switch Spectrum 16A QS16	No.	2	R	R
	Install				
2.12.7	16 A , 1-lever , 1-way - 50x100,	No.	8	R	R
2.12.8	16 A , 1-lever , 1x2-way - 50x100,	No.	2	R	R
2.12.9	16 A , 1-lever and Rotary dimmer - Shuttle - 50x100,	No.	1	R	R
2.12.10	16 A , 1-lever, 1x1way, Weatherproof - IP65	No.	1	R	R
2.12.11	Recessed single-load 360deg PIR occupancy sensor. Schneider Electric: SAE-UE-MS-CSBWE	No.	14	R	R
2.12.12	Photocell, 16A, Daylight Switch Spectrum 16A QS16	No.	2	R	R
2.13	Plug sockets for luminaires				
	Supply				
2.13.1	5A 3pin socket outlet, unswitched, for power connection to luminaire, 60mm diameter.	No	100	R	R
	Install				
2.13.2	5A 3pin socket outlet, unswitched, for power connection to luminaire, 60mm diameter.	No	0	R	R
2.14	Luminaires				
	Supply, install, connect and commission the following luminaires and as per the specification luminaire schedule.				
	Supply				
2.14.1	Type A Light Fitting	No	41	R	R
2.14.2	Type B Light Fitting	No	16	R	R
Carried Forward to next Page					R

BILL: ELECTRICAL INSTALLATION

Item	Description	Unit	Qty	Rate	Amount
Brought Forward from previous Page					R
2.14.3	Type C1 Light Fitting	No	1	R	R
2.14.4	Type C2 Light Fitting	No	1	R	R
2.14.5	Type C3 Light Fitting	No	1	R	R
2.14.6	Type C4 Light Fitting	No	1	R	R
2.14.7	Type D Light Fitting	No	34	R	R
2.14.8	Type E Light Fitting	No	2	R	R
2.14.9	Type F Light Fitting	No	1	R	R
2.14.10	Type G Light Fitting	No	5	R	R
2.14.11	Type H Light Fitting	No	2	R	R
2.14.12	Type T Light Fitting	No	4	R	R
Install					
2.14.13	Type A Light Fitting	No	41	R	R
2.14.14	Type B Light Fitting	No	16	R	R
2.14.15	Type C1 Light Fitting	No	1	R	R
2.14.16	Type C2 Light Fitting	No	1	R	R
2.14.17	Type C3 Light Fitting	No	1	R	R
2.14.18	Type C4 Light Fitting	No	1	R	R
2.14.19	Type D Light Fitting	No	34	R	R
2.14.20	Type E Light Fitting	No	2	R	R
2.14.21	Type F Light Fitting	No	1	R	R
2.14.22	Type G Light Fitting	No	5	R	R
2.14.23	Type H Light Fitting	No	2	R	R
2.14.24	Type T Light Fitting	No	4	R	R
Carried Forward to next Page					R

BILL: ELECTRICAL INSTALLATION

Item	Description	Unit	Qty	Rate	Amount
Brought Forward from previous Page					R
2.15	Earthing installation				
2.15.1	Supply and install the project main earthing installation complete with all necessary equipment e.g. electrodes, copper conductors, draw boxes, terminations on main earth bar etc. to comply with Supply Authority requirements and the Specification.	Sum	1	R	R
2.15.2	Bonding of taps and hand wash basins.	Sum	1	R	R
2.15.3	Bonding of gutters and down pipes	Sum	1	R	R
2.15.4	Earthing continuity tests.	Sum	1	R	R
2.15.5	Testing and commissioning of project main earthing installation to the satisfaction of the Supply Authority. (including earth resistance tests.)	Sum	1	R	R
2.16	Uninterrupted Power Supply (UPS)				
2.16.1	Supply, install and commission a 5KVA, 230V on-line UPS with 30 minute battery backup time at full load	Sum	1	R	R
Sub total for Section 2 - General Electrical					R

BILL: ELECTRICAL INSTALLATION

3	Sum Values and Allowances				
3.1	Allow for the following provisional amounts:				
3.1.1	Allowance to obtain the services of an accredited Access Control supplier and installer as selected by the engineer, Principal Agent and Client. Carry to Amount.	Prov. Sum	R	96 250,00	R
3.1.2	Profit and attendance on item 3.1.1 above	%	R	96 250,00	R
3.1.3	Allowance to obtain the services of an accredited CCTV supplier and installer as selected by the engineer, Principal Agent and Client. Carry to Amount.	Prov. Sum	R	69 850,00	R
3.1.4	Profit and attendance on item 3.1.3 above	%	R	69 850,00	R
3.1.5	Allowance to obtain the services of an accredited Data cabling supplier and installer as selected by the engineer, Principal Agent and Client. Carry to Amount	Prov. Sum	R	174 900,00	R
3.1.6	Profit and attendance on item 3.1.5 above	%	R	174 900,00	R
3.1.7	Allowance for Design Refinement. Carry to Amount	Prov. Sum	R	25 000,00	R
3.1.8	Profit and attendance on item 3.1.7 above	%	R	25 000,00	R
	Note: Written quotes from suppliers will be required for these items, as for proof of cost.				
Sub total for Section 3 - Sum Values and Allowances					R

BILL: ELECTRICAL INSTALLATION

SUMMARY

Sub total for Section1 - Preliminary and General Items R

Sub total for Section 2 - General Electrical R

Sub total for Section 3 - Sum Values and Allowances R

Sub Total R

15% VAT R

Total R

Particulars of specialist Electrical Contractor:

Name of specialised Electrical Contractor:

As on the registration certificate as an electrical contractor, from the Department of Labour, as per the

Registered Number :

Name of installation electrician on registration certificate:

Registration number of the above person:

INFORMATION SCHEDULE

Item	Description	Trademark/Manufacturer
1	Cable Tray, Wire basket and trunking	
2	Power Skirting	
3	PVC Conduit and accessories	
4	Galvanised conduit and accessories.	
5	PVC-insulated conductors	
6	Light Fittings - Provide Lighting Schedule	
7	Plugs and switches	

On behalf of the Bidder, who warrants his/her authority

Contact Person:

Cell Phone Number:

E-Mail Address:

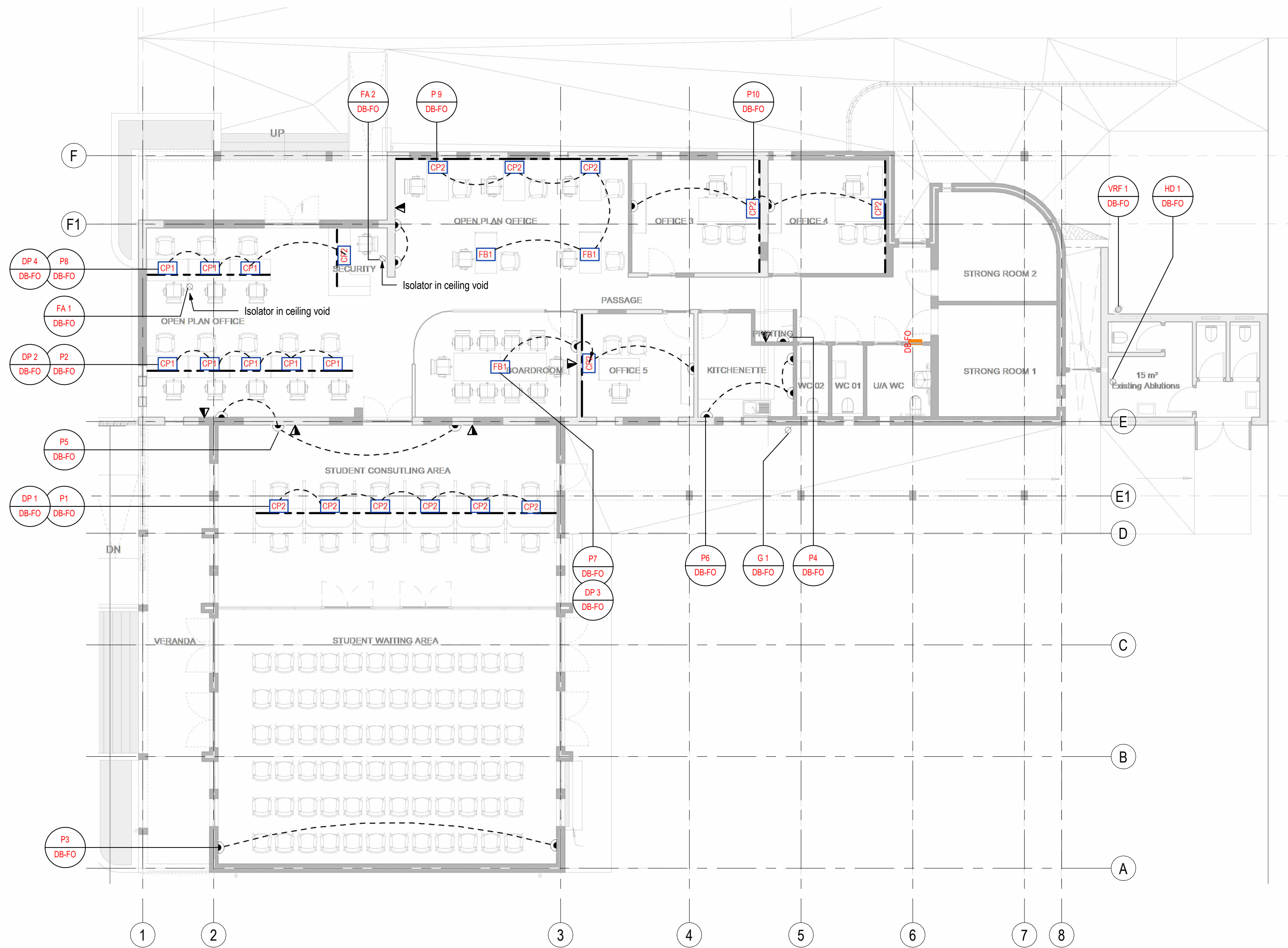
Signature of Electrical contractor:

Date:

6. ANNEXURES

6.1 DRAWING REGISTER

DRAWING NO.	TYPE	DESCRIPTION
P8080E01-001	A1	ELECTRICAL INSTALLATION: POWER LAYOUT
P8080E01-002	A1	ELECTRICAL INSTALLATION: LIGHTING LAYOUT
P8080E01-003	A3	ELECTRICAL INSTALLATION: DISTRIBUTION BOARD LAYOUT



2 GROUND FLOOR POWER
1 : 100

LEGEND	
	DISTRIBUTION BOARD
	DUO 16A SSO - 1xOld Type 1xNew Type
	SET ON OUTLETS IN POWER SKIRTING CONSISTING OF: 1 x OLD RSA SSO, 1 x NEW RSA SSO(EURO), 2xDED SSO, 2xRJ45 OUTLETS & COMBO SOCKET OUTLET
	SET ON OUTLETS IN POWER SKIRTING CONSISTING OF: 1xOLD RSA SSO, 1xNEW RSA SSO(EURO), 2xDED SSO, 1xUSB & 2xRJ45 OUTLETS
	FLOOR MOUNTED PEDESTAL PLUG COMPLETE WITH: 1xOLD RSA SSO, 1xNEW RSA SSO(EURO)
	DATA POINT (RJ45)
POWER SKIRTING, 2 TIER, PVC, COMPLETE WITH BENDS AND END CAPS AS REQUIRED. TO BE MOUNTED AT HEIGHTS AS PER THE ARCHITECTS DETAIL TO SUITE JOINERY.	
LUMINAIRES	
	REC - 600X600 - 2 CELL DIFFUSER
	ROUND BULKHEAD, BLACK, 15 W
	55w x 80h LINEAR LIGHT, ±3.6m LONG
	55w x 80h LINEAR LIGHT, ±5.4m LONG
	55w x 80h LINEAR LIGHT, ±3m LONG
	55w x 80h LINEAR LIGHT, ±6m LONG
	DOWNLIGHT, RECESSED
	FOOTLIGHT, ± 2W
	Ø1200mm ROUND LED
	UPLIGHT GU10 FITTING
	COLOUR TEMPERATURE LED FLOOD LIGHT
	SPLASH PROOF LED FITTING
	LIGHT SWITCH, 16A, 1 WAY
	LIGHT SWITCH, 16A, 2WAY
	MOTIONPRESENCE DETECTOR
	PHOTOCELL, DAYLIGHT SWITCH

No.	Date	Description
A	2025/11/06	Issued for information.
B	2025/11/10	Move lighting to sheet P8099E01-002.
C	2025/12/15	Additional SSO and Data points as shown. ISSUED FOR TENDER

NOTE:
ALL MEASUREMENTS TO BE CHECKED ON SITE.
IF IN DOUBT, ASK.
THIS DESIGN & DRAWING IS THE INTELLECTUAL & CONFIDENTIAL PROPERTY OF CA du Toit EC (PTY) LTD AND MAY NOT BE DISCLOSED TO THIRD PARTY, COPIED OR REPRODUCED IN WHOLE OR IN PART WITHOUT THE PERMISSION OF CA du Toit EC (PTY) LTD.

CAduToit
CONSULTING ENGINEERS / RAADGEWENDE INGENIEURS

Eastern Cape (Pty) Ltd
22 Shirley Street
Newton Park
GOEBERHA
6045

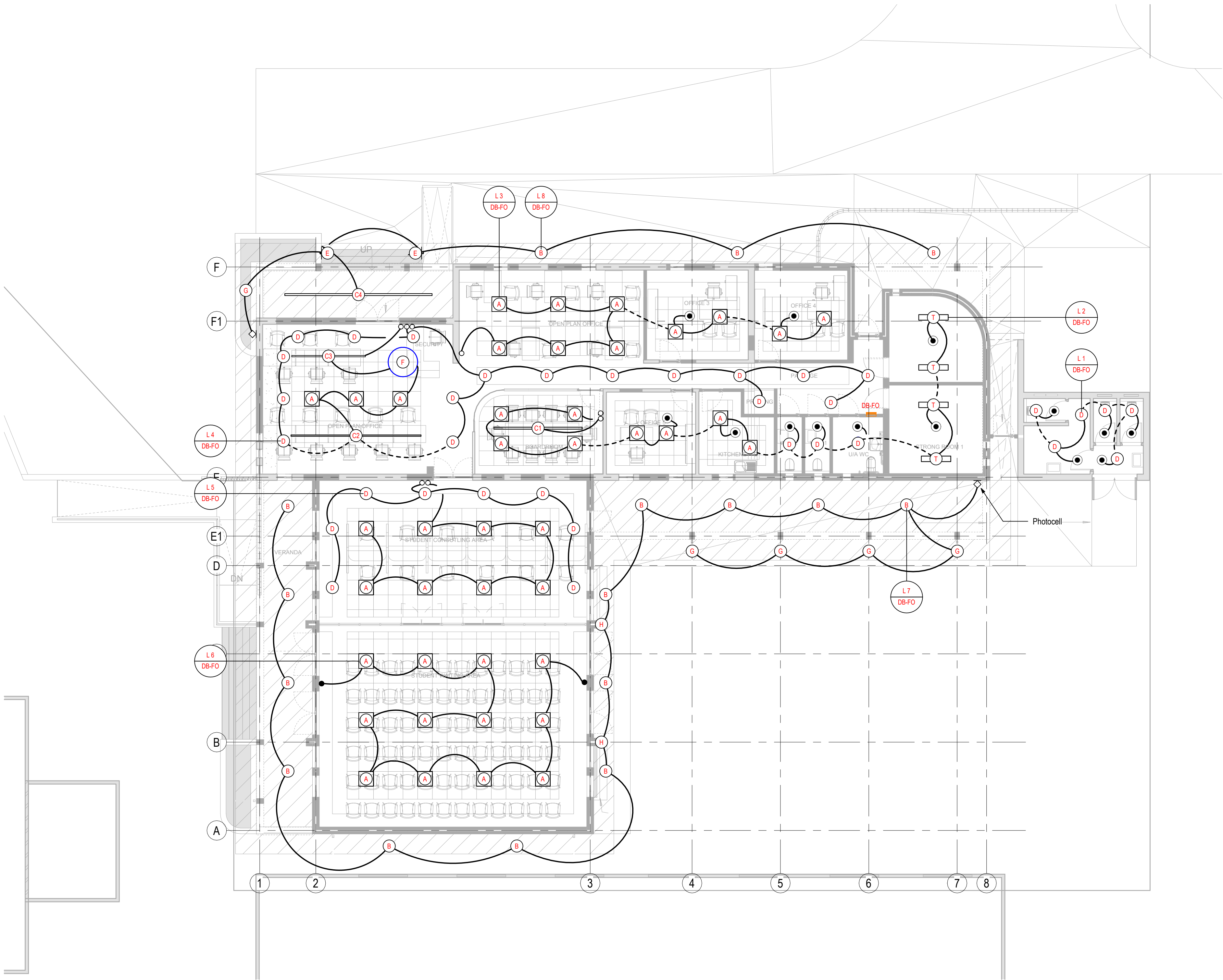
E-mail
pe@cadutoitec.co.za
Telephone
+27 (0)41 585 7559

PROJECT	FORT HARE UNIVERSITY FINANCE OFFICES
DRAWING	ELECTRICAL INSTALLATION POWER LAYOUT

SCALE (on A1):	DATE:
1 : 100	JULY 2025
DRAWN:	DESIGNED:
C. CLAASEN	L. BRIGHT
APPROVED:	VERIFIED:
L. BRIGHT	P. STRYDOM

ISSUED FOR:	TENDER
-------------	--------

DRAWING NUMBER:	Project Ref:	P8099E01
P8099E01-001		
Revision No.:		
C		



GROUND FLOOR LIGHTING

1 : 100

LEGEND	
	DISTRIBUTION BOARD
	DUO 16A SSO - 1xOld Type 1xNew Type
	SET ON OUTLETS IN POWER SKIRTING CONSISTING OF: 1 x OLD RSA SSO, 1 x NEW RSA SSO(EURO), 2xDED SSO, 2xRJ45 OUTLETS & COMBO SOCKET OUTLET
	SET ON OUTLETS IN POWER SKIRTING CONSISTING OF: 1xOLD RSA SSO, 1xNEW RSA SSO(EURO), 2xDED SSO, 1xUSB & 2xRJ45 OUTLETS
	FLOOR MOUNTED PEDESTAL PLUG COMPLETE WITH: 1xOLD RSA SSO 1xNEW RSA SSO(EURO)
	DATA POINT (RJ45)
POWER SKIRTING, 2 TIER, PVC, COMPLETE WITH BENDS AND END CAPS AS REQUIRED, TO BE MOUNTED AT HEIGHTS AS PER THE ARCHITECTS DETAIL TO SUITE JOINERY.	
LUMINAIRES	
	REC - 600x600 - 2 CELL DIFFUSER
	ROUND BULKHEAD, BLACK, 15 W
	55w x 80h LINEAR LIGHT, ±3.6m LONG
	55w x 80h LINEAR LIGHT, ±5.4m LONG
	55w x 80h LINEAR LIGHT, ±3m LONG
	55w x 80h LINEAR LIGHT, ±6m LONG
	DOWNLIGHT, RECESSED
	FOOTLIGHT, ± 2W
	Ø1200mm ROUND LED
	UPLIGHT GU10 FITTING
	COLOUR TEMPERATURE LED FLOOD LIGHT
	SPLASH PROOF LED FITTING
	LIGHT SWITCH, 16A, 1 WAY
	LIGHT SWITCH, 16A, 2WAY
	MOTION/PRESENCE DETECTOR
	PHOTOCELL, DAYLIGHT SWITCH

No.	Date	Description
A	2025/11/10	Revised lighting to architect comments.
B	2025/12/12	ADDED CIRCUITS
C	2025/12/15	ISSUED FOR TENDER

NOTE:
ALL MEASUREMENTS TO BE CHECKED ON SITE.
IF IN DOUBT, ASK.
THIS DESIGN & DRAWING IS THE INTELLECTUAL & CONFIDENTIAL PROPERTY OF CA du Toit EC (PTY) LTD AND MAY NOT BE DISCLOSED TO THIRD PARTY, COPIED OR REPRODUCED IN WHOLE OR IN PART WITHOUT THE PERMISSION OF CA du Toit EC (PTY) LTD.

CAduToit Eastern Cape (Pty) Ltd
CONSULTING ENGINEERS / RAADGEWENDE INGENIEURS
22 Shirley Street
Newton Park
GOEBERHA
6045
E-mail
pe@cadutoilec.co.za
Telephone
+27 (0)41 585 7559

PROJECT	FORT HARE UNIVERSITY FINANCE OFFICES
DRAWING	ELECTRICAL INSTALLATION LIGHTING LAYOUT

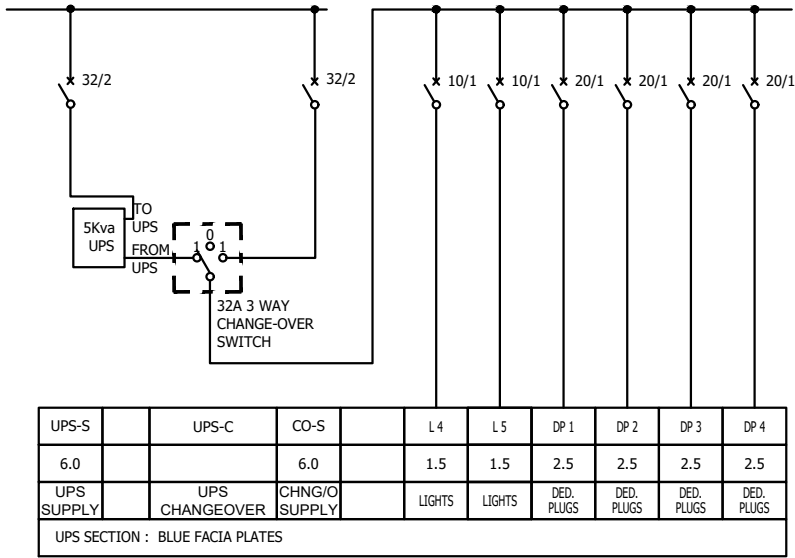
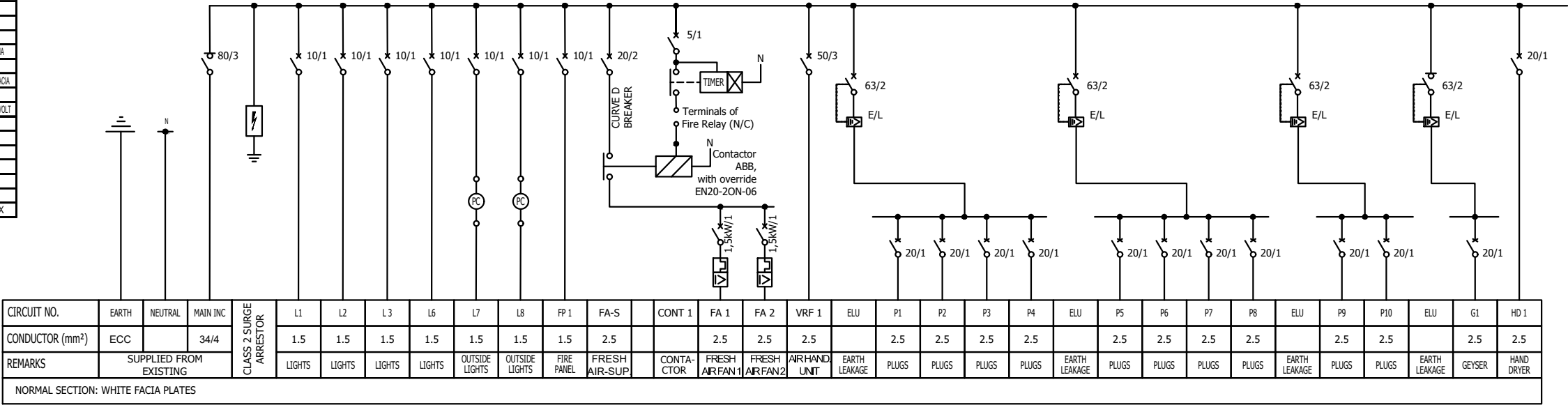
SCALE (on A1):	DATE:
1 : 100	JULY 2025
DRAWN:	DESIGNED:
C. CLAASEN	L. BRIGHT
APPROVED:	VERIFIED:
L. BRIGHT	P. STRYDOM

ISSUED FOR:	TENDER
-------------	--------

DRAWING NUMBER:	Project Ref:	P8099E01
P8099E01-002		Revision No.:
		C

DB-FO

DISTRIBUTION BOARD INFORMATION				
CONSTRUCTION	YES	NO	LIMITING DIMENSIONS	
RECESSED			WIDTH:	MOS
SEMI-RECESSED	X		HEIGHT:	MOS
SURFACE			DEPTH:	MOS
FREE STANDING			COLOR SABS 1081	
FLOOR STANDING			STANDBY SECTION (E)	WHITE FACIA
DOORS			UPS SECTION (U)	BLUE FACIA
DOORS LOCKABLE	X		ISOLATED TX SECTION (IS)	WHITE FACIA
SEPARATE CUBICLES				
Spare Space	35%	%	VOLTAGE:	400 VOLT
SERVICE ACCESS	FRONT:	X	REAR:	BOTH:
ENTRY:	TOP:	X	BOTTOM:	SIDE:
INCOMING:		X		
OUTGOING:		X		X
BUSBAR RATING	RATING (N)		RATING (E)	FAULT LEVEL
PHASE	3 PHASE	80		5 KA
BUSBAR NEUTRAL			HALF:	FULL: X



SHEET SIZE

A3

REV	DATE	DESCRIPTION
A	2025/12/15	ISSUED FOR TENDER

NOTE:
ALL MEASUREMENTS TO BE CHECKED ON SITE.
IF IN DOUBT, ASK.
THIS DESIGN & DRAWING IS THE INTELLECTUAL & CONFIDENTIAL PROPERTY OF **CA du Toit EC (PTY) LTD** AND MAY NOT BE DISCLOSED TO THIRD PARTY, COPIED OR REPRODUCED IN WHOLE OR IN PART WITHOUT THE PERMISSION OF **CA du Toit EC (PTY) LTD**.

CAduToit
CONSULTING ENGINEERS / RAADGEWENDE INGENIEURS

22 Shirley Street
Newton Park
Gqeberha
6045

E-mail
pe@cadutoitec.co.za
Telephone
+27 (0)41 585 7559

PROJECT:
**FORT HARE UNIVERSITY
FINANCE OFFICES**

DRAWING:
DISTRIBUTION BOARD LAYOUT

SCALE: N. T. S.	DATE: 2025/12/15
DRAWN: C. CLAASEN	DESIGNED: P. STRYDOM
APPROVED: P. STRYDOM	VERIFIED: L. BRIGHT

ISSUED FOR:
TENDER

CONSULTANTS JOB REFERENCE NUMBER.
P8080E01-003

SHEET NO:
REV. NO: **A**